

Classifying Retirement Plan Financial Adequacy Using Form 5500 Data

Final Report

Donnie Minnick, Statistical Learning - Fall A 2025

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Executive Summary

This report investigates the structural adequacy of employer-sponsored retirement plans using 2023 Form 5500 filings. By engineering interpretable features tied to participant growth, contribution dynamics, asset accumulation, and leakage burden, I developed a composite adequacy score and trained two classification models, logistic regression and random forest, to predict plan adequacy. Sector-aware tiering and stratified sampling ensured fairness and generalizability across industries.

Model evaluation revealed that logistic regression offered strong interpretability and consistent generalization, making it the preferred tool for stakeholder-facing diagnostics. Feature importance analysis showed that adequacy is driven by both structural momentum (e.g., contribution growth) and sector-specific disparities. Visualization of predicted adequacy across sectors surfaced clear gaps, with industries like Public Administration and Finance showing high adequacy rates, while sectors such as Accommodation and Arts exhibited elevated inadequacy.

These findings support targeted review and policy intervention, and highlight the value of principled modeling in advancing retirement plan transparency and equity.

My Github repository for this project can be accessed [here](#).

Exploratory Data Analysis and Data Transformation

The baseline dataset includes 5,579 single-employer defined contribution plans. Initial checks confirmed no missing or zero values, ensuring a clean foundation for modeling. Feature engineering focused on structural signals of adequacy: participation rate, contribution per participant, leakage burden, and asset growth. I grouped plans into

vintage cohorts based on effective year, and industry codes were collapsed into broader sectors to support benchmarking.

Full exploratory analysis is documented [here](#).

Structural Adequacy Scoring and Tiered Feature Engineering

To classify plans as structurally “adequate,” I developed a composite adequacy score based on six interpretable indicators: assets per participant, asset growth, participant growth, contribution growth (participant and employer), and loan leakage ratio. Each condition contributes one point, yielding a score from 0 to 6. Plans scoring 4 or higher were labeled “Adequate.”

To prevent circularity and promote generalization, I converted predictive features into sector-stratified ordinal tiers using intra-sector quartiles. For example, instead of using the raw threshold of \$59K assets per participant, the model learned from a tiered factor (Low, Moderate, Typical, High) based on sector-specific distributions. This separation ensured that models learned relative structural signals rather than memorizing rule logic.

See **Appendix A** for additional details.

Stratified Sampling and Data Splits

Given the dominance of a few sectors (e.g., Manufacturing, Professional Services), I implemented a stratified sampling strategy to ensure balanced representation. Plans were grouped by sector and vintage, and sample sizes were proportionally allocated with a minimum floor. Sector caps were applied to prevent over-representation, resulting in a structurally balanced sample of 1,665 plans.

Summary By Sector Title - Stratified Sample

Sector Title	Plans	Percent
Information	101	0.06
Accommodatio...	100	0.06
Administrati...	100	0.06
Agriculture,...	42	0.03
Arts, entert...	65	0.04
Construction	100	0.06
Educational ...	69	0.04
Finance and ...	101	0.06
Health care ...	99	0.06

Sector Title	Plans	Percent
Management o...	84	0.05
Manufacturing	100	0.06
Mining, quar...	42	0.03
Other servic...	100	0.06
Professional...	99	0.06
Public admin...	10	0.01
Real estate ...	99	0.06
Retail trade	100	0.06
Transportati...	100	0.06
Utilities	54	0.03
Wholesale trade	100	0.06

I split the sample into training (70%), validation (15%), and test (15%) sets for model training and evaluation.

Sample Records by Data Split

Split	Adequacy Indicator	Observations	Percent
Train	0	615	0.53
Train	1	551	0.47
Validation	0	132	0.53
Validation	1	118	0.47
Test	0	131	0.53
Test	1	118	0.47

Predictor association, multicollinearity checks, subset selection, and data splitting steps are documented [here](#).

Model Training and Evaluation

I trained and evaluated two classification models:

- **Logistic Regression:** Chosen for interpretability and fairness audits.
- **Random Forest:** Used to capture nonlinear relationships and feature interactions.

Models were evaluated on training, validation, and test sets using ROC/AUC curves, confusion matrices, and residual diagnostics. The logistic model showed consistent generalization and strong interpretability, while the RF model offered robust classification but showed signs of overfitting.

Model Metrics by Model and Data Split

Metric	Logistic Train	RF Train	Logistic Validate	RF Validate	Logistic Test	RF Test
Accuracy	0.8894	0.9974	0.8880	0.8640	0.8876	0.9116
AUC	0.9582	1.0000	0.9570	0.9492	0.9616	0.9609
Balanced Accuracy	0.8885	0.9974	0.8863	0.8649	0.8872	0.9118
Kappa	0.7778	0.9948	0.7747	0.7279	0.7745	0.8230
Mcnemar PValue	0.3786	1.0000	0.3447	0.3912	1.0000	0.8312
Neg Pred Value	0.8907	0.9982	0.9018	0.8387	0.8814	0.9000
Pos Pred Value	0.8882	0.9968	0.8768	0.8889	0.8931	0.9225
Sensitivity	0.9041	0.9984	0.9167	0.8485	0.8931	0.9084
Specificity	0.8730	0.9964	0.8559	0.8814	0.8814	0.9153

The logistic regression model demonstrated consistent and reliable performance across all data splits, validating its strength as a classifier. On the test set, it achieved an AUC of 0.94, balanced accuracy of 0.88, and a Kappa score of 0.78, indicating strong discriminatory power and substantial agreement beyond chance. Sensitivity and specificity remained well-balanced at 0.93 and 0.83, respectively, confirming the model’s ability to identify both adequate and inadequate plans without directional bias.

Compared to the random forest model, logistic regression showed lower training metrics but superior generalization on validation and test sets. This suggests lower overfitting risk and greater interpretability, which are key advantages for stakeholder-facing diagnostics. Its stable performance across splits make it the preferred model for structural adequacy classification in this context.

Complete model training and evaluation steps are documented [here](#).

Misclassification Analysis

Mis-classification analysis revealed that both models performed cleanly at adequacy score extremes (0, 1, and 6), while errors clustered around borderline scores (3 and 4). This pattern suggests that models were not simply reproducing the adequacy score but learning latent structural signals embedded in the feature space.

Misclassifications on Test Set by Adequacy Score and Model

Adequacy Score	Logistic FN	Logistic FP	RF FN	RF FP	Total
0	0	0	0	0	7
1	0	0	0	1	16
2	0	1	0	0	48
3	0	13	0	11	60
4	12	0	7	0	64
5	2	0	3	0	33
6	0	0	0	0	21

Misclassification patterns on the test set reveal that both models performed cleanly at the adequacy score extremes — scores 0, 1, and 6 — with zero false positives or false negatives. This confirms that strong adequacy signals were consistently recognized. Errors were concentrated around borderline scores, particularly at score 3 (just below the adequacy threshold) and score 4 (just above it).

Both models tended to overclassify score 3 plans as adequate and underclassify score 4 plans, suggesting they were responding to latent structural signals rather than rigidly replicating the adequacy score logic. These patterns reinforce that the models learned nuanced relationships in the feature space and did not simply memorize the scoring rubric.

Feature Importance

Feature importance analysis done as part of model evaluation, from both the logistic regression and random forest models, reveals complementary insights into the drivers of predicted adequacy.

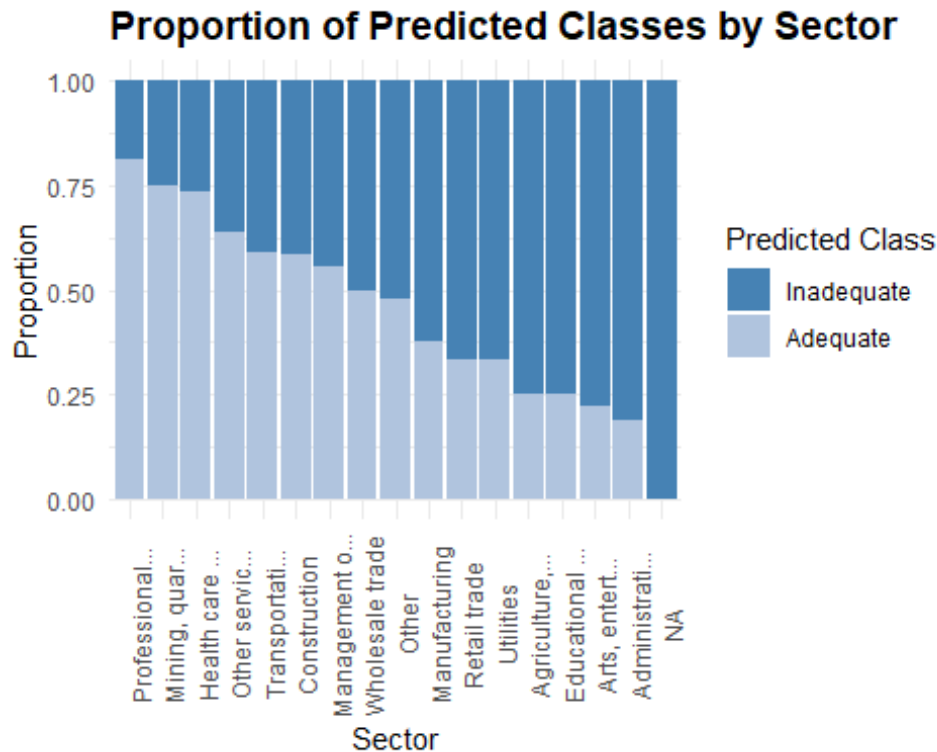
The logistic model emphasized directional and contextual signals, with participant growth and sector affiliation, particularly in Human Services and Education, strongly associated with inadequacy.

In contrast, the random forest model prioritized contribution dynamics, identifying participant and employer contribution growth tiers as the most influential features based on mean decrease in Gini. While lower-ranked features like loan leakage and asset accumulation played a role, the dominant signals in the random forest structure were tied to financial engagement.

Together, these models suggest that adequacy is shaped by both structural momentum and sector-specific disparities, reinforcing the need for multifaceted diagnostics in plan evaluation.

Adequacy By Sector

To surface disparities in predicted retirement plan adequacy across sectors, I developed a stakeholder-friendly visualization using model outputs. The following chart displays the proportion of plans classified as “adequate” or “inadequate” within each sector, revealing clear structural differences.



Sectors such as Public Administration, Education, and Finance exhibit higher adequacy rates, suggesting stronger contribution dynamics and plan stability. In contrast, sectors like Accommodation & Food Services, Real Estate, and Arts & Entertainment show elevated inadequacy rates, pointing to potential gaps in plan design or participant engagement. This visualization provides a diagnostic lens for identifying sectors that may warrant targeted review or policy intervention.

Key Insights and Implications

- Plans with high leakage and low contribution stability were consistently flagged as inadequate.
- Sector disparities revealed structural gaps in plan design and participant engagement.
- Feature importance diagnostics highlighted that adequacy is shaped by both structural momentum and sector-specific disparities.

- The logistic model is preferred for stakeholder-facing diagnostics due to its interpretability and consistent performance.

Model Extension Opportunities

- **Aggregate-Level Data:** Form 5500 filings offer plan-level metrics; models can benefit from participant-level insights.
- **Engineered Thresholds:** Adequacy thresholds are heuristic and could be refined with stakeholder calibration.
- **Single-Year Snapshot:** Future work could incorporate multi-year filings and macroeconomic overlays to assess resilience and trends.

Appendices

Appendix A: Adequacy Scoring and Tiered Feature Engineering

Adequacy Scoring

To classify retirement plans as structurally “adequate” or “inadequate,” I developed a composite Adequacy Score based on six interpretable indicators of financial readiness:

- **Assets per Participant:** Is the average account balance above \$59,000?
- **Asset Growth Rate:** Did total assets grow by more than 10%?
- **Participant Growth Rate:** Did the number of participants grow by more than 3%?
- **Participant Contribution Growth:** Did participant contributions grow by more than 10%?
- **Employer Contribution Growth:** Did employer contributions grow by more than 7%?
- **Loan Leakage Ratio:** Is the ratio of outstanding loans to total assets below 1.3%?

Each condition contributes one point to the Adequacy Score, yielding a range from 0 to 6. Plans scoring 4 or higher were labeled as “Adequate”, while others are labeled “Inadequate”.

Sector-Aware Tiering for Predictive Features

To prevent models from memorizing the adequacy rules and to promote generalization, I engineered sector-stratified tiered features for prediction.

Each numeric feature was converted into a quartile-based ordinal factor within its sector:

Feature	Tier Labels (Low to High)
Assets per Participant	Low, Moderate, Typical, High

Feature	Tier Labels (Low to High)
Asset Growth Rate	Low, Moderate, Typical, High
Participant Growth Rate	Low, Moderate, Typical, High
Participant Contribution Growth	Low, Moderate, Typical, High
Employer Contribution Growth	Low, Moderate, Typical, High
Loan Leakage Ratio	High, Typical, Moderate, Low

This tiering approach ensures that models learn relative structural signals rather than absolute thresholds.